

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1.(10 points) Suppose that, in each period, the cost of a security either goes up by a factor of 2 or goes down by a factor of $1/2$. Assume the nominal interest rate per period is $r = 10\%$.

(a) Compute the risk-neural probabilities $(p^*, 1 - p^*)$.

$$\gamma = 0.1, u = 2, d = \frac{1}{2}$$
$$\Rightarrow p^* = \frac{1 + \gamma - d}{u - d} = \frac{1 + 0.1 - 0.5}{2 - 0.5} = \frac{0.6}{1.5} = 0.4$$
$$1 - p^* = 1 - 0.6 = 0.4$$

(b) Suppose that the initial price of the security is 100, determine the no-arbitrage cost of a call option to purchase the security at the end of 2 periods for a strike price of 80.

$S=100, \quad t=2, \quad K=80$

A binomial tree diagram showing stock price evolution over two time steps. The root node is 100. At time 1, the price can go up to 200 or down to 50. At time 2, from 200 it can go up to 400 or down to 100, and from 50 it can go up to 100 or down to 25. The tree is drawn with nodes at each time step and branches labeled 'u' for up and 'd' for down.

$$C_{\text{kur}} = (400 - 80)(1+r)^{-2} = \frac{320}{(1,1)^2} \approx 264.46$$

$$C_{nd} = C_{du} = (100 - 80)(4\gamma)^{-2} = \frac{20}{(0.11)^2} \approx 16.53$$

$$C_{dd} = 0$$

Thus

$$\begin{aligned} C &= P_{uu} C_{uu} + P_{ud} C_{ud} + P_{du} C_{du} + P_{dd} C_{dd} \\ &= (p^*)^2 C_{uu} + 2 p^* (1-p^*) C_{ud} + (1-p^*)^2 \cdot C_{dd} \\ &= (0.4)^2 (264.46) + 2 \times 0.4 \times 0.6 \times 16.53 + 0 \\ &= 50.25 \end{aligned}$$